

Network Models (OSI Reference Model)

Delivered by

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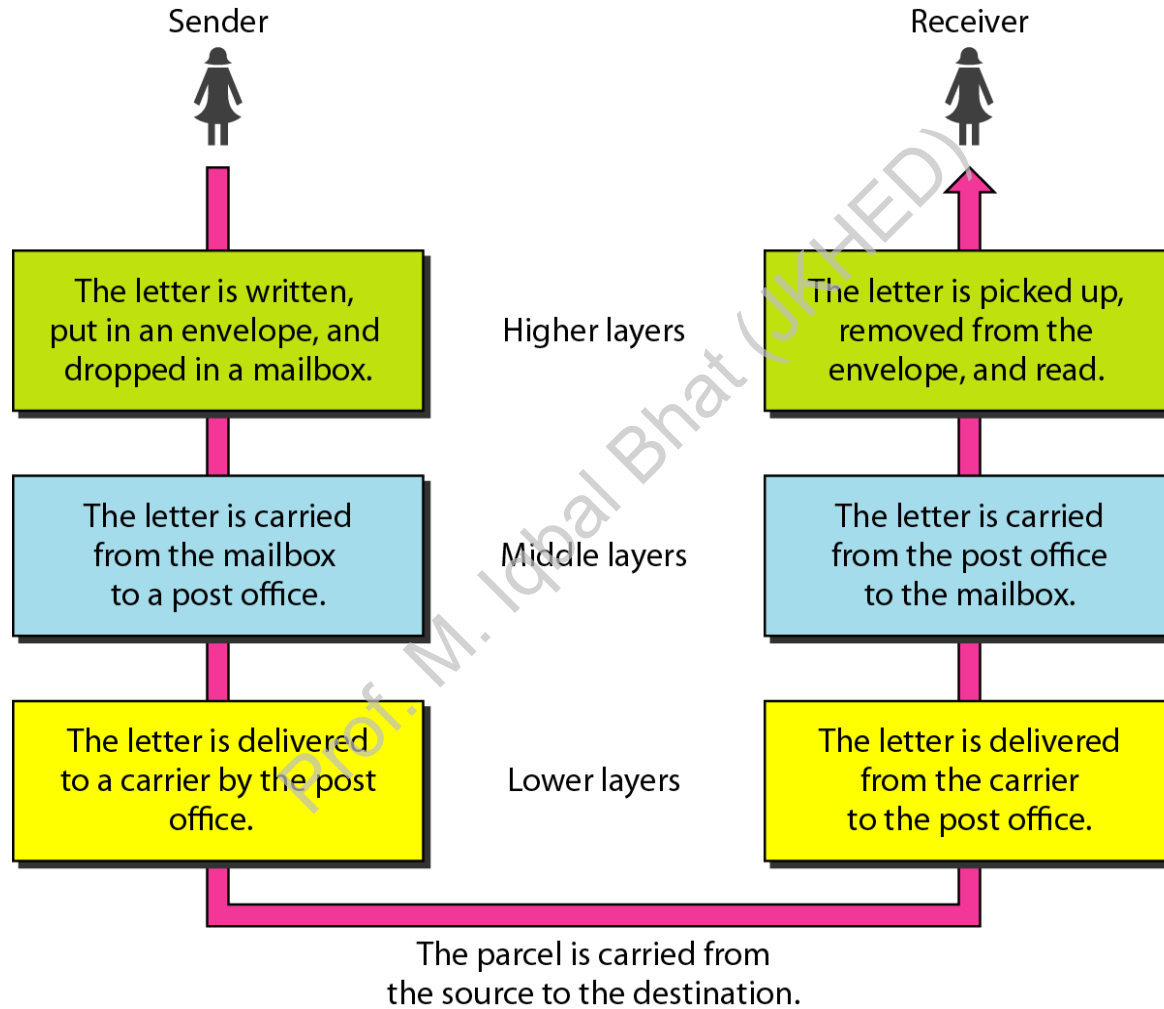
2-1 LAYERED TASKS

*We use the concept of **layers** in our daily life. As an example, let us consider two friends who communicate through postal mail. The process of sending a letter to a friend would be complex if there were no services available from the post office.*

Topics discussed in this section:

**Sender, Receiver, and Carrier
Hierarchy**

Figure 2.1 Tasks involved in sending a letter



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OSI Model

2-2 THE OSI MODEL

*Established in 1947, the International Standards Organization (**ISO**) is a multinational body dedicated to worldwide agreement on international standards. An ISO standard that covers all aspects of network communications is the Open Systems Interconnection (**OSI**) model. It was first introduced in the late 1970s.*

Topics discussed in this section:

Layered Architecture

Peer-to-Peer Processes

Encapsulation



Note

**ISO is the organization.
OSI is the model.**

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Figure 2.2 *Seven layers of the OSI model*

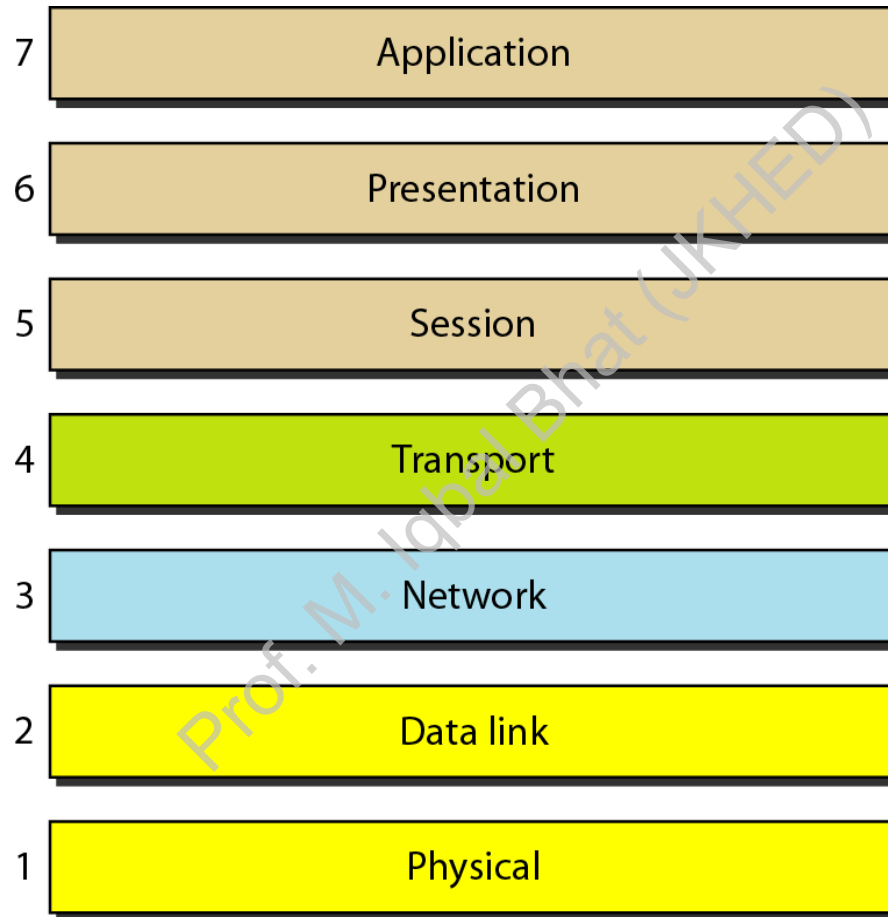


Figure 2.3 *The interaction between layers in the OSI model*

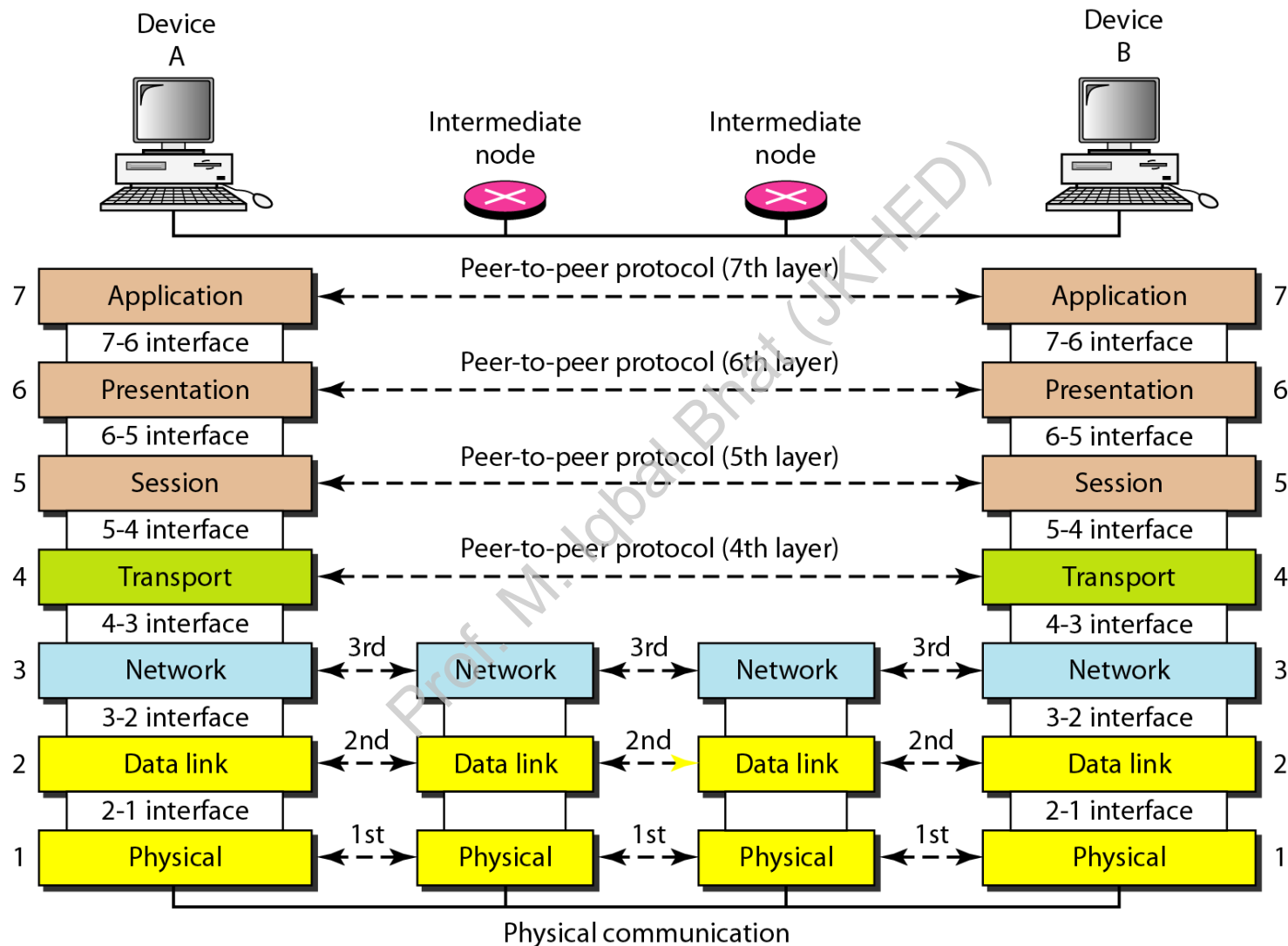
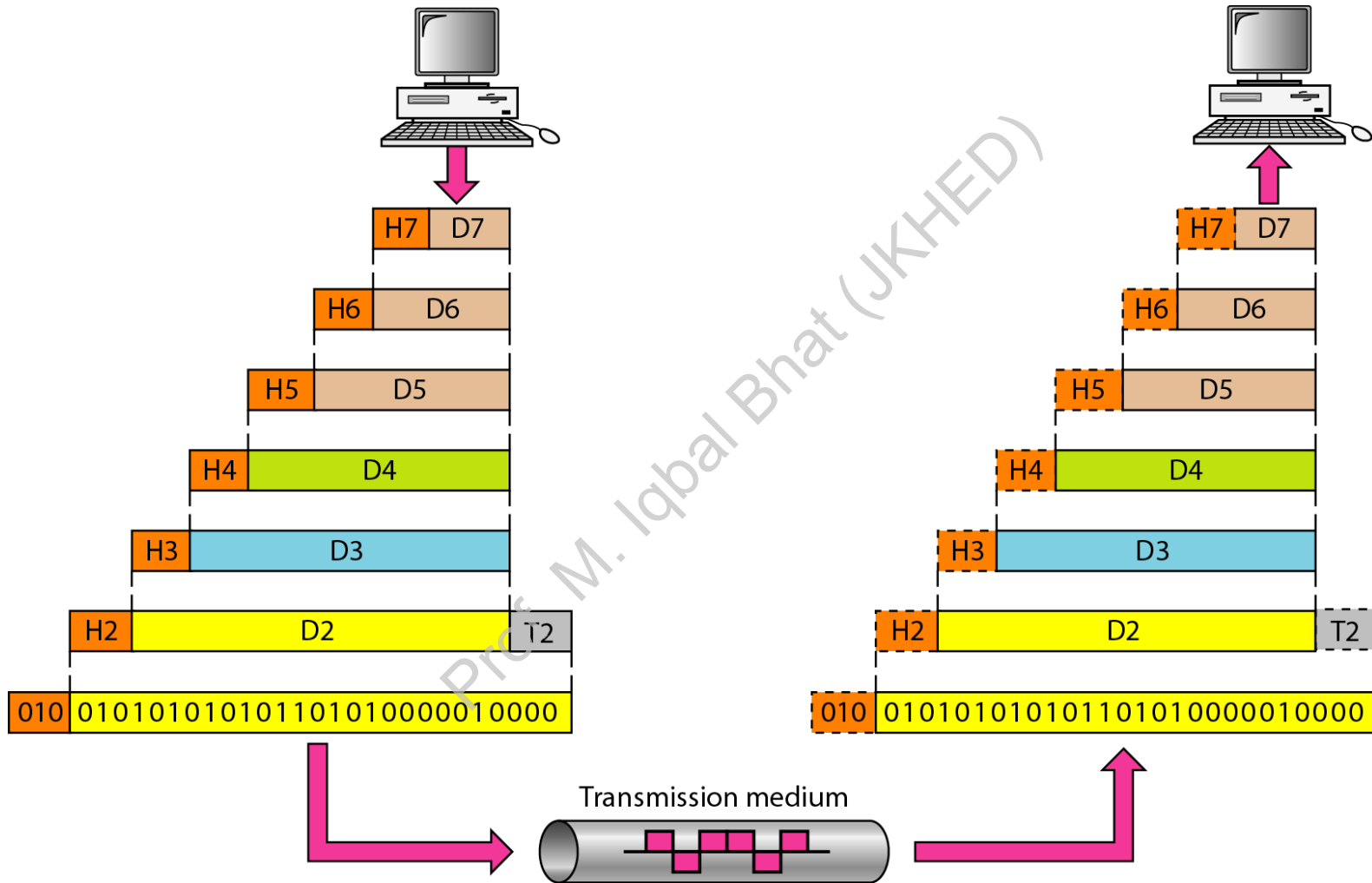


Figure 2.4 *An exchange using the OSI model*



2-3 LAYERS IN THE OSI MODEL

In this section we briefly describe the functions of each layer in the OSI model.

Topics discussed in this section:

Physical Layer

Data Link Layer

Network Layer

Transport Layer

Session Layer

Presentation Layer

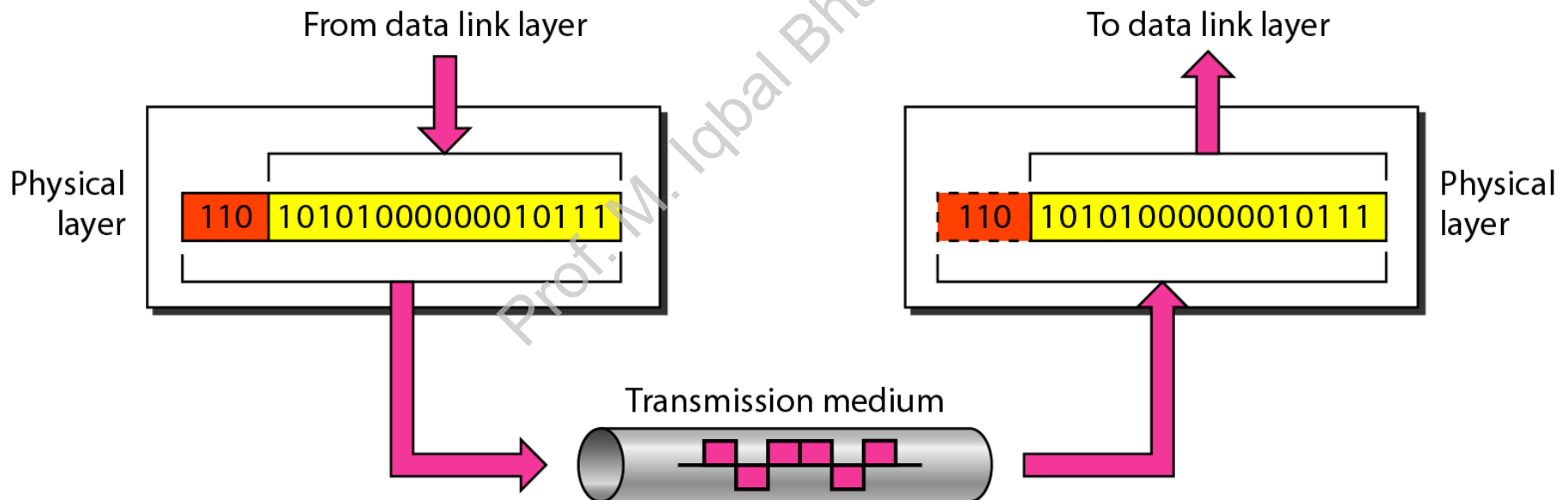
Application Layer

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Figure 2.5 *Physical layer*

What it does: Deals with the physical connection between devices.

Analogy: Think of it as the actual cables, wires, and hardware (like network cards) that connect your computer to a network.





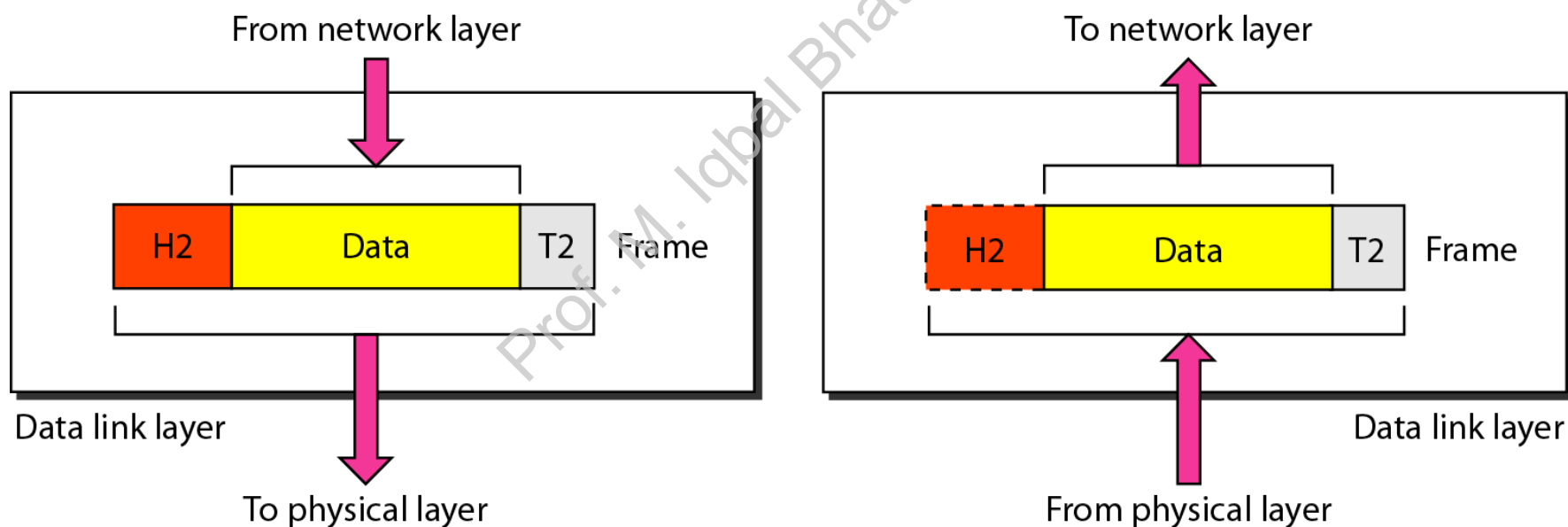
Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Figure 2.6 *Data link layer*

What it does: Ensures reliable point-to-point and point-to-multipoint communication.

Analogy: Imagine it as the traffic cop that makes sure data gets from one point to another without collisions.



Note

The data link layer is responsible for moving frames from one hop (node) to the next.

Figure 2.7 *Hop-to-hop delivery*

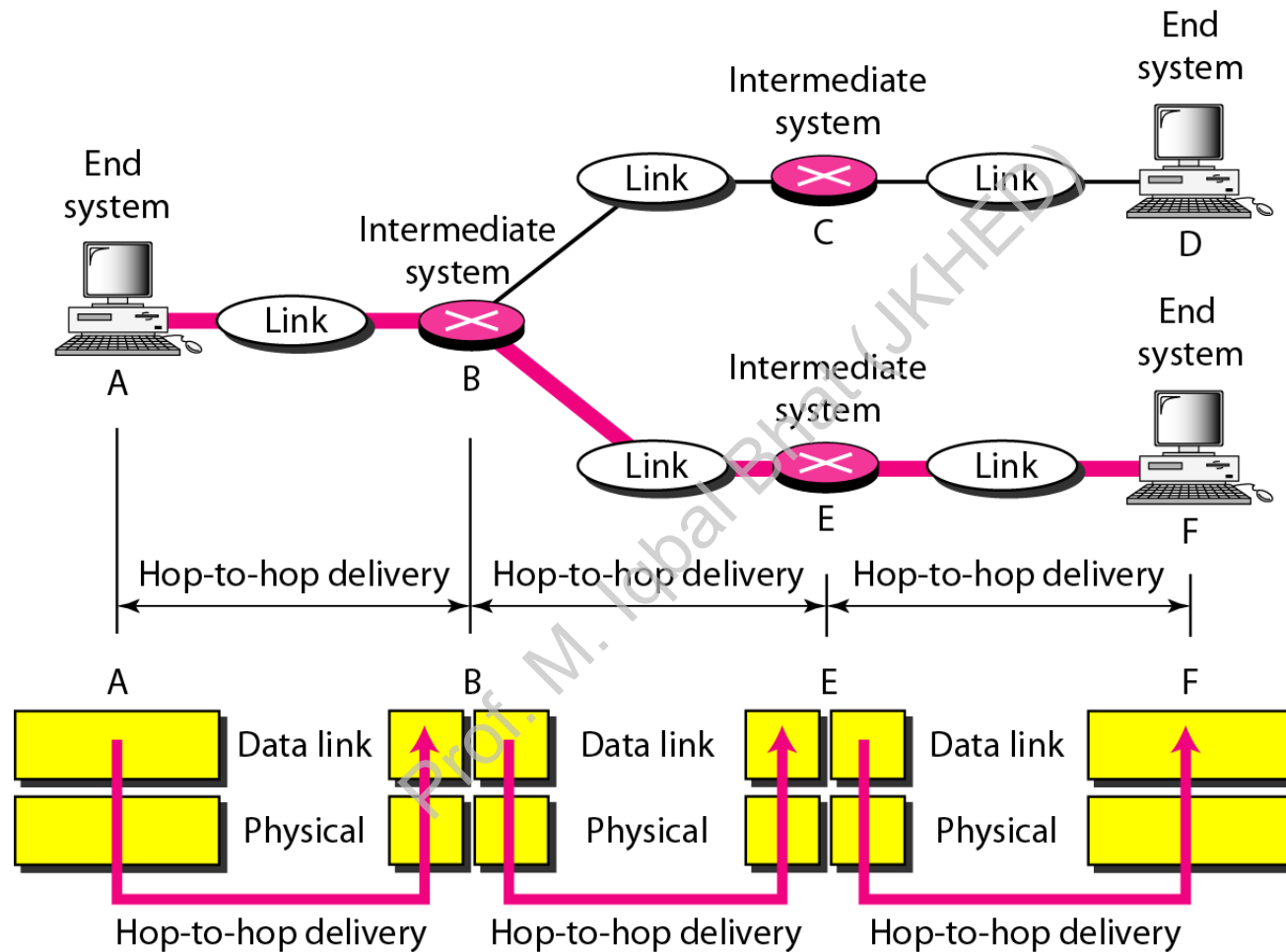
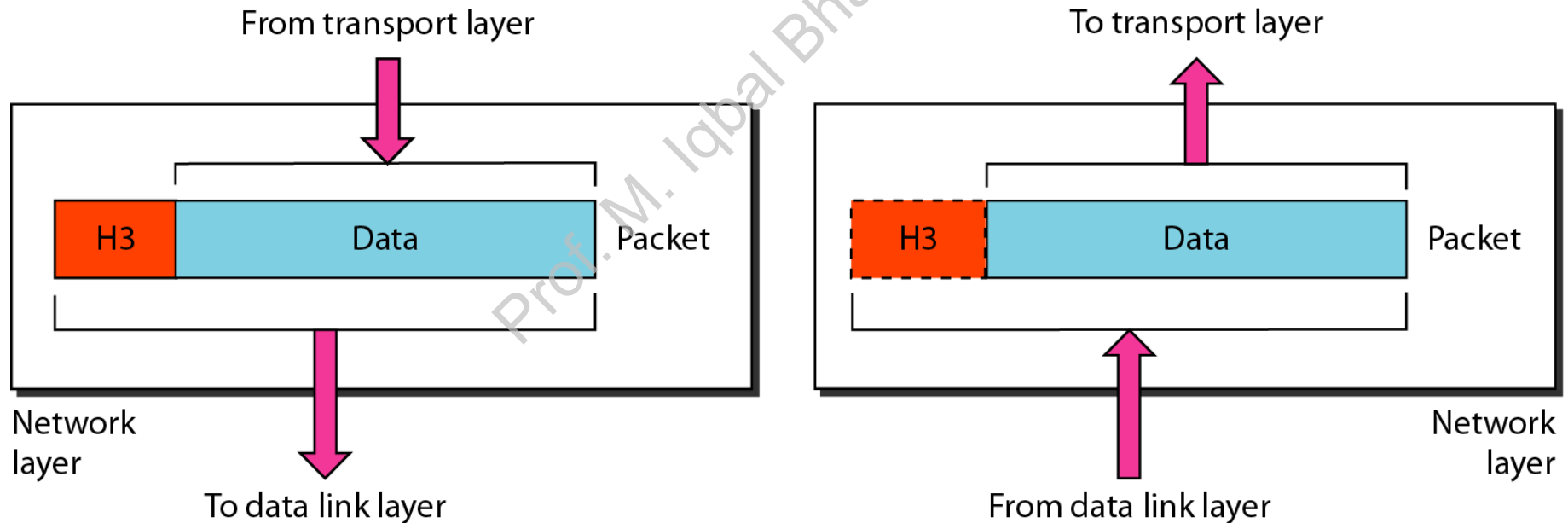


Figure 2.8 *Network layer*

What it does: Manages the addressing and routing of data between devices on a network.

Analogy: Like a postal system, it puts addresses on data and decides the best route for it to reach its destination.



Note

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Figure 2.9 *Source-to-destination delivery*

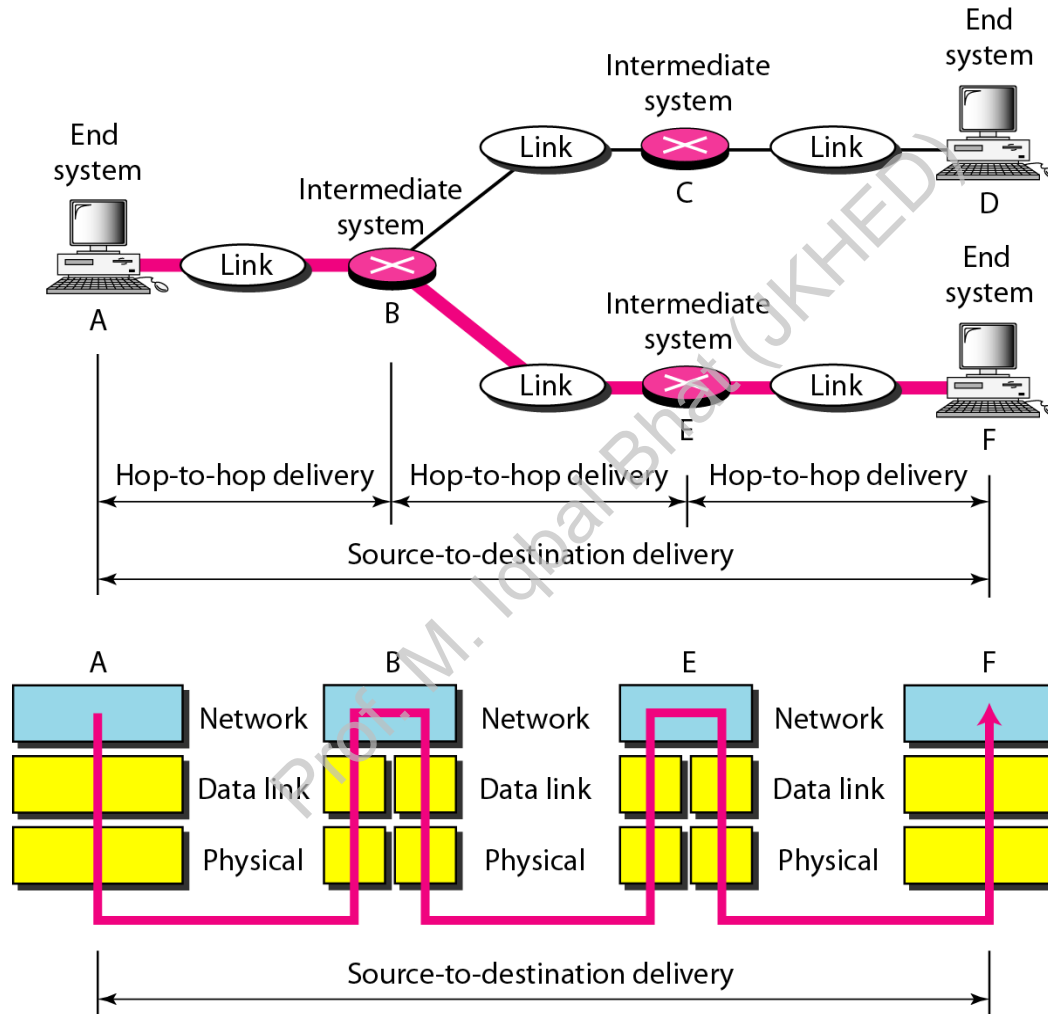


Figure 2.10 *Transport layer*

What it does: Ensures end-to-end communication and error-checking.

Analogy: Similar to a conversation, it manages how much information to send and ensures it arrives intact.

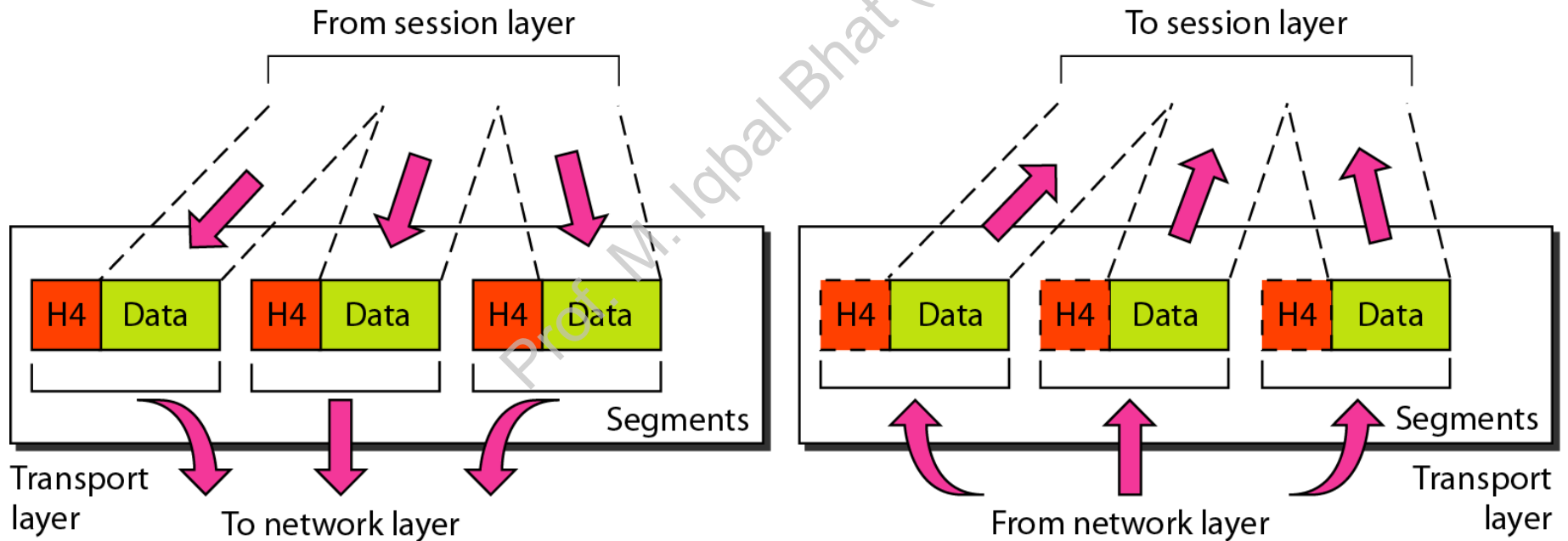
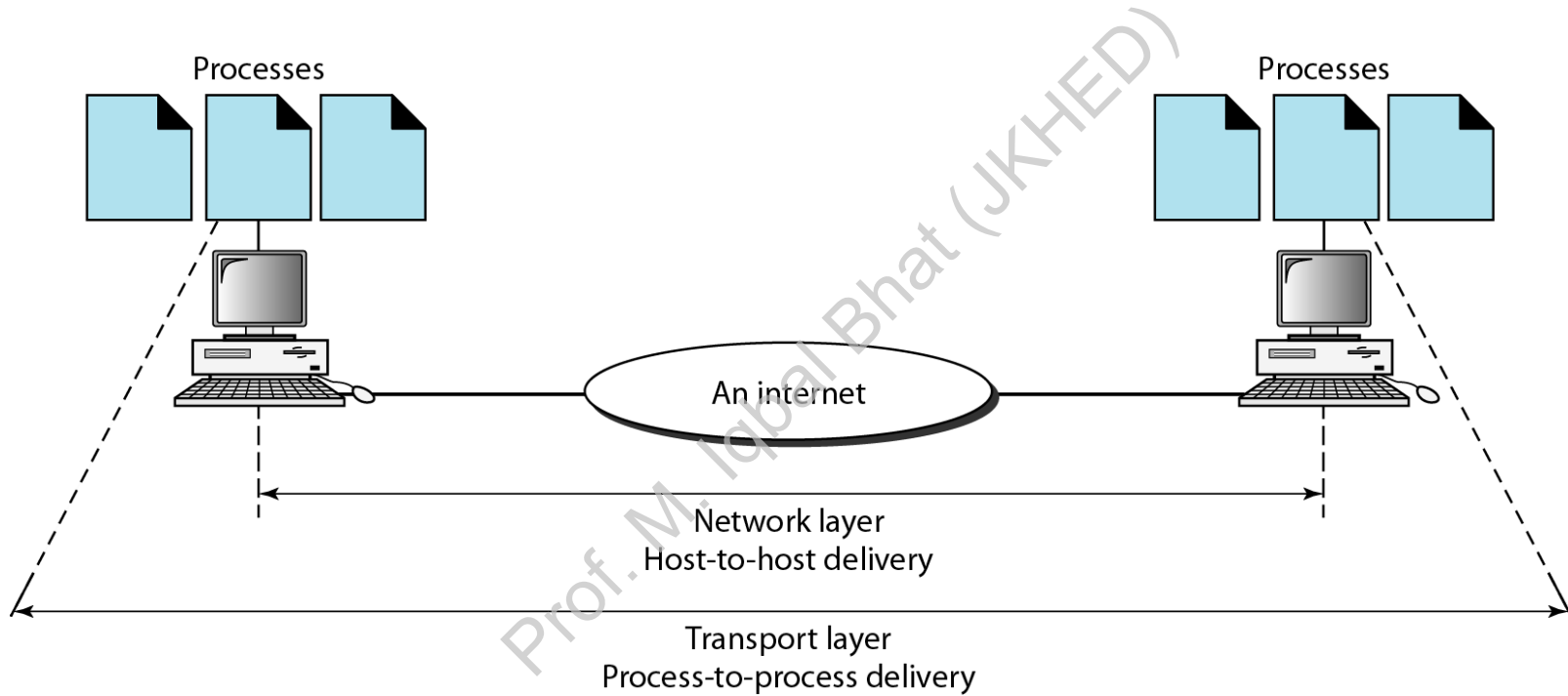


Figure 2.11 *Reliable process-to-process delivery of a message*



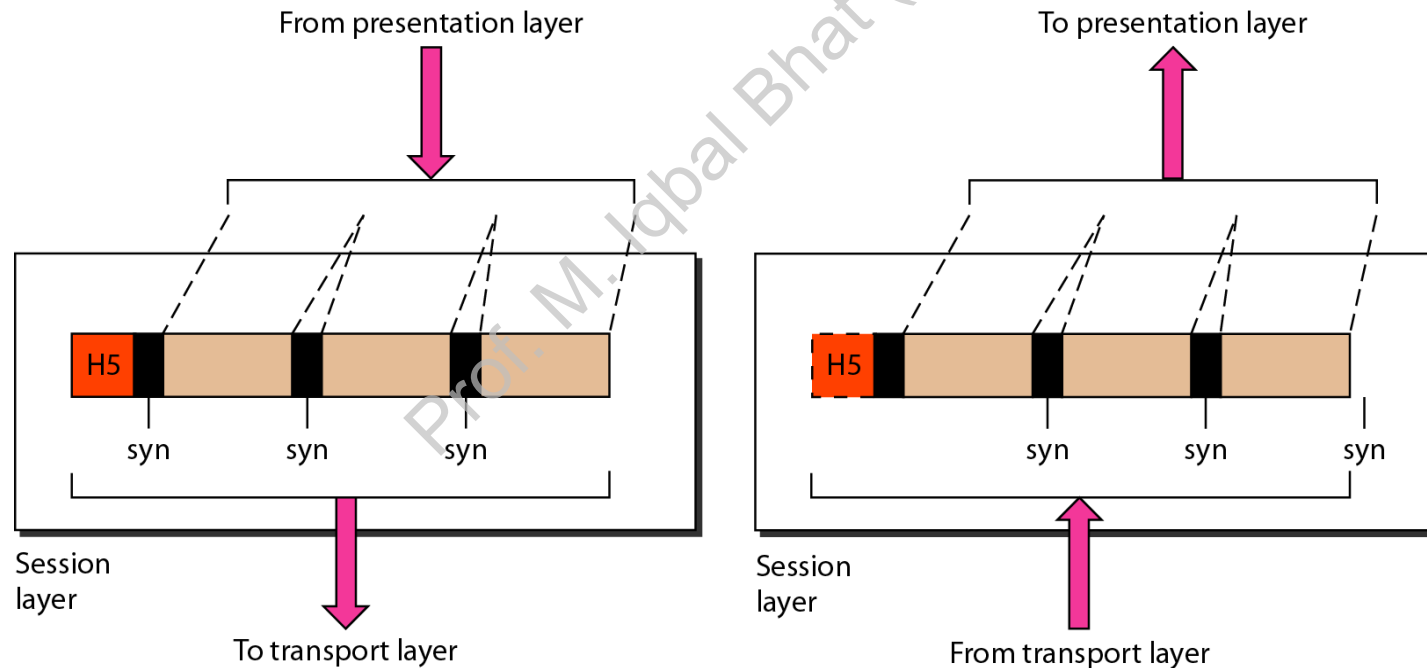
Note

The transport layer is responsible for the delivery of a message from one process to another.

Figure 2.12 *Session layer*

What it does: Establishes, maintains, and terminates connections (sessions) between applications.

Analogy: Think of it as the manager of a phone call, deciding when to start and end the conversation.



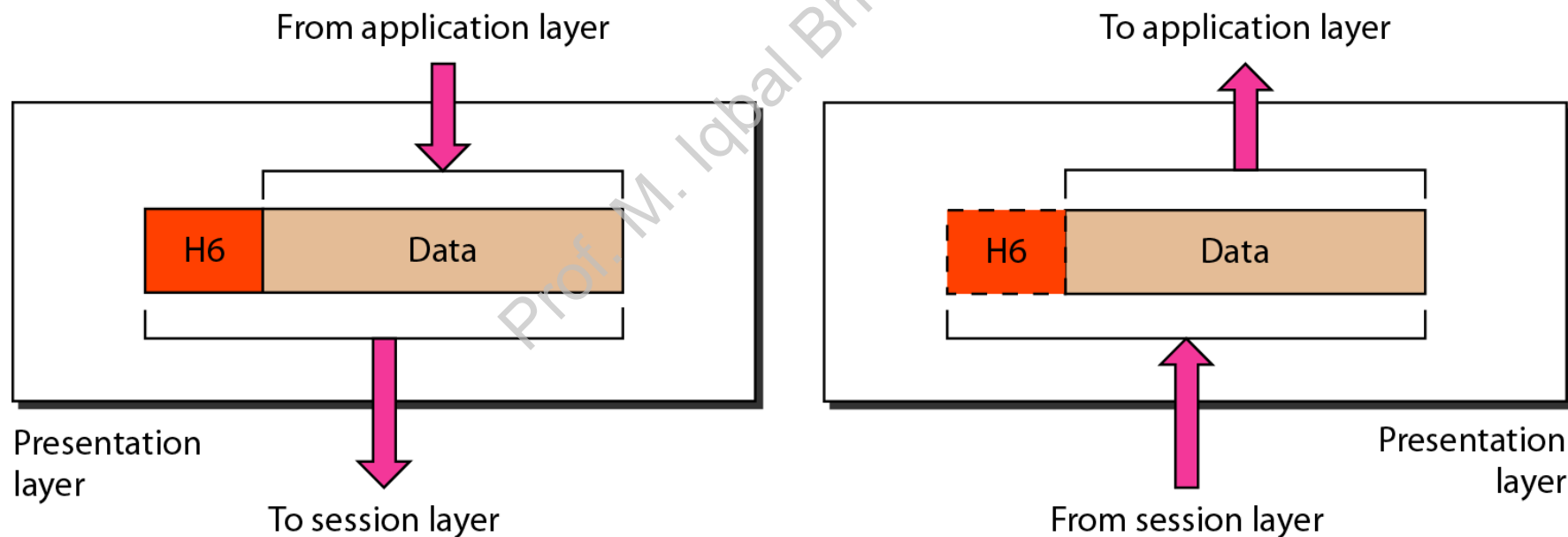
Note

The session layer is responsible for dialog control and synchronization.

Figure 2.13 *Presentation layer*

What it does: Translates data between the application layer and the lower layers.

Analogy: Like a translator at a United Nations meeting, making sure everyone understands the data being sent.





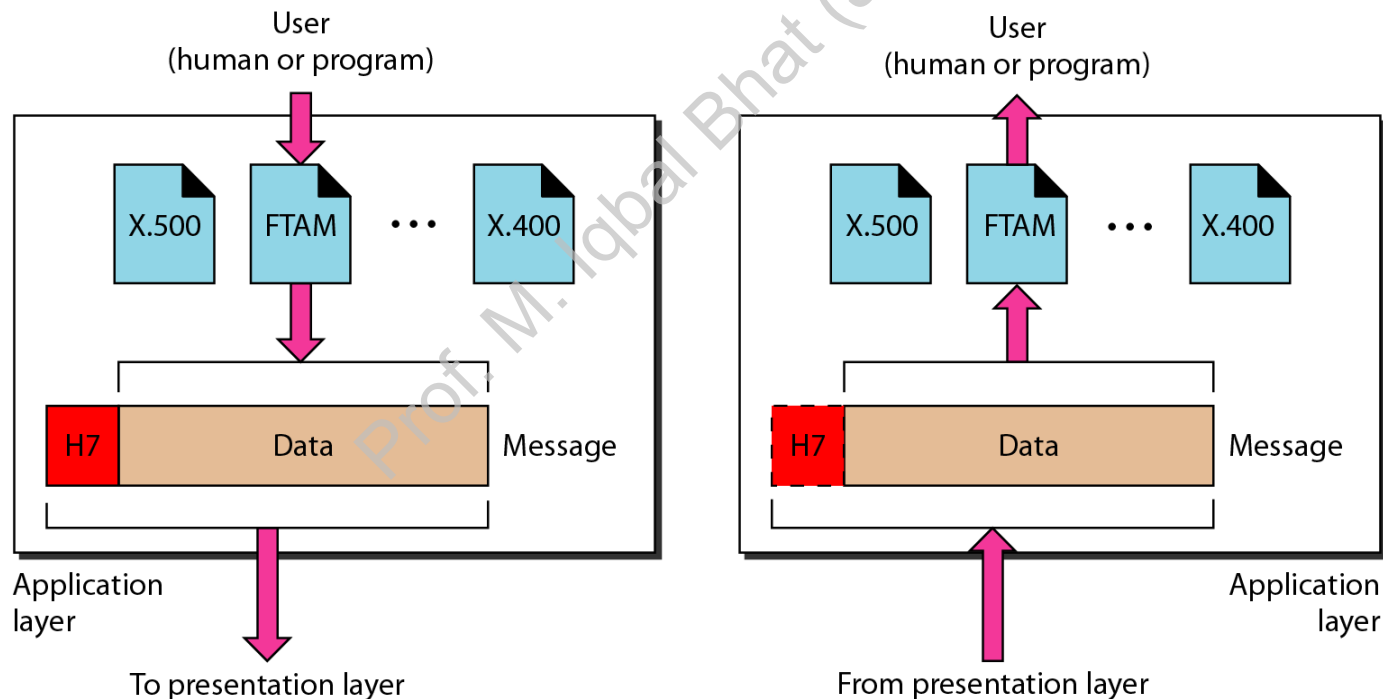
Note

The presentation layer is responsible for translation, compression, and encryption.

Figure 2.14 *Application layer*

What it does: Provides network services directly to end-users or applications.

Analogy: This is the layer you interact with - it's like the software you use for email, web browsing, etc.

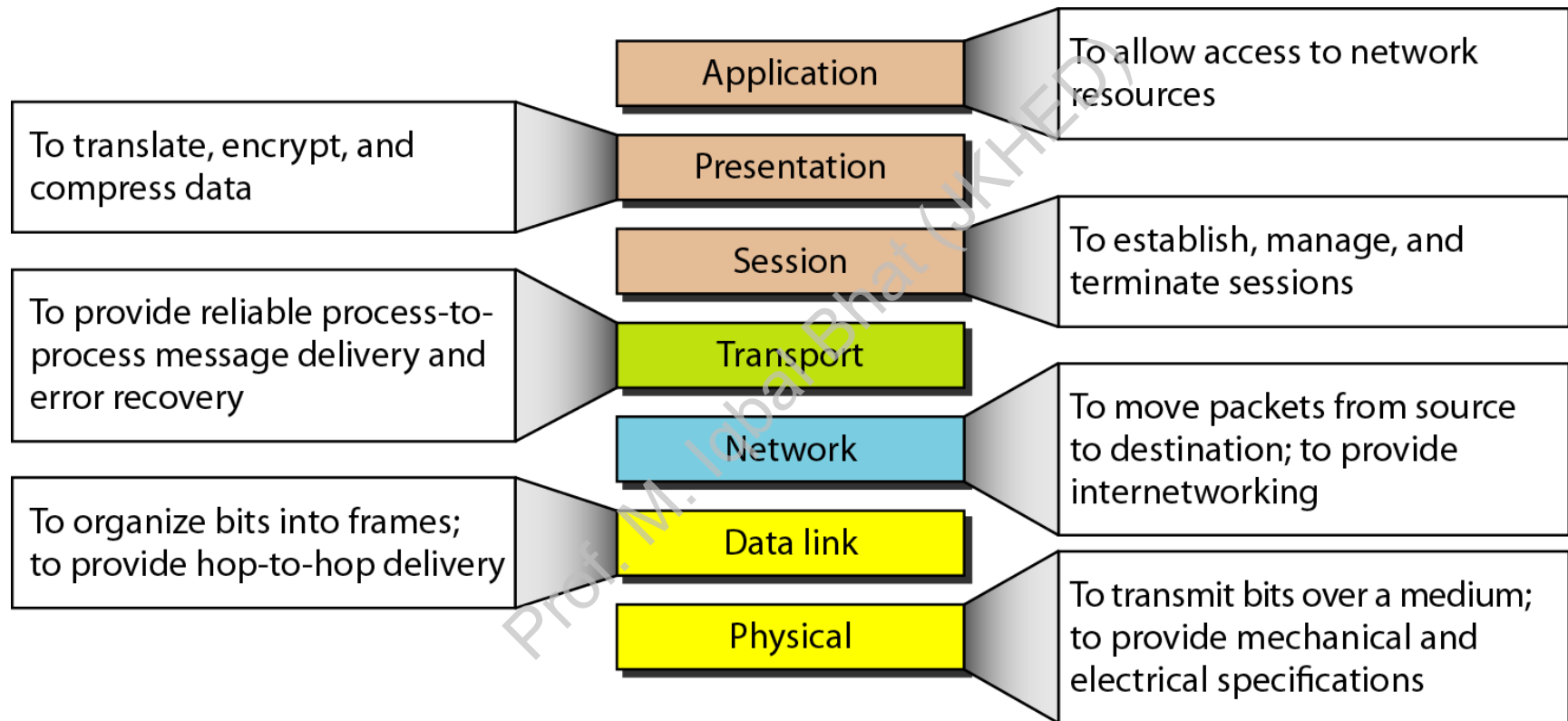




Note

The application layer is responsible for providing services to the user.

Figure 2.15 *Summary of layers*



2-4 TCP/IP PROTOCOL SUITE

The layers in the **TCP/IP protocol suite** do not exactly match those in the OSI model. The original TCP/IP protocol suite was defined as having four layers: **host-to-network**, **internet**, **transport**, and **application**. However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers: **physical**, **data link**, **network**, **transport**, and **application**.

Topics discussed in this section:

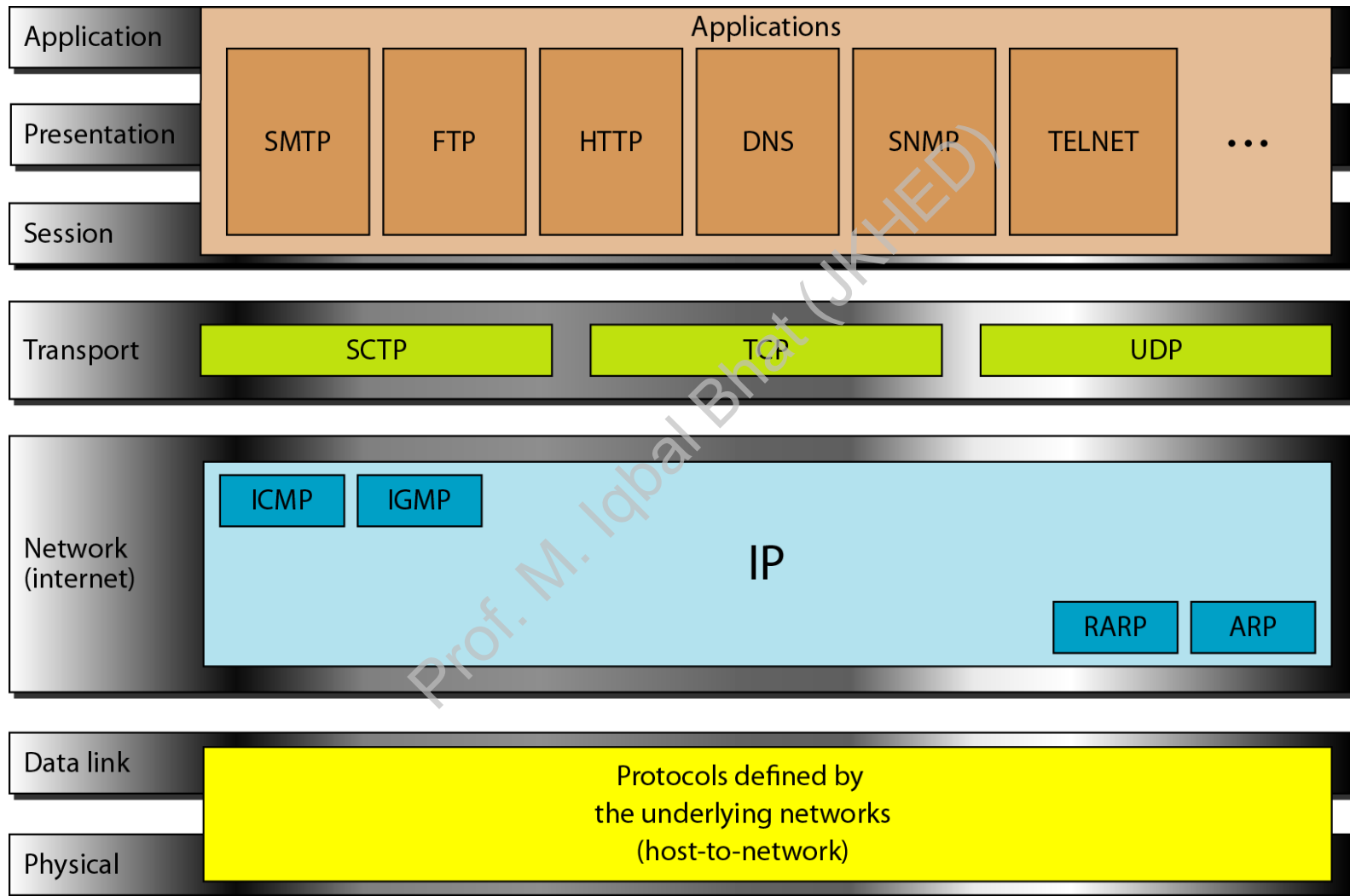
Physical and Data Link Layers

Network Layer

Transport Layer

Application Layer

Figure 2.16 *TCP/IP and OSI model*



1. Host-to-Network (Physical Layer)

What it does: Similar to the Data Link Layer in the OSI model, it handles the physical connection to the network and is concerned with the local delivery of frames between devices on the same network.

Analogy: Think of it as the address on an envelope that ensures a letter gets to the right house in your neighborhood.

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2. Internet Layer

What it does: Corresponds to the Network Layer in the OSI model. It is responsible for addressing, routing, and fragmentation and reassembly of packets for end-to-end communication across multiple networks.

Analogy: Similar to the postal service, it decides the best route for your letter to travel across different neighborhoods and cities.

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3. Transport Layer

What it does: Functions similarly to the Transport Layer in the OSI model. It ensures reliable, end-to-end communication and may include error checking and flow control mechanisms.

Analogy: Think of it like a conversation between two people where they decide how much information to exchange and ensure that it's received correctly.

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4. Application Layer

What it does: Similar to the Application Layer in the OSI model, it provides communication services directly to end-users and applications. It contains various protocols for specific applications like HTTP for web browsing, FTP for file transfer, and SMTP for email.

Analogy: This is the layer where you interact with network services and applications, like using a web browser, sending emails, or downloading files.

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2-5 ADDRESSING

*Four levels of addresses are used in an internet employing the TCP/IP protocols: **physical**, **logical**, **port**, and **specific**.*

Topics discussed in this section:

Physical Addresses

Logical Addresses

Port Addresses

Specific Addresses

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Figure 2.17 *Addresses in TCP/IP*

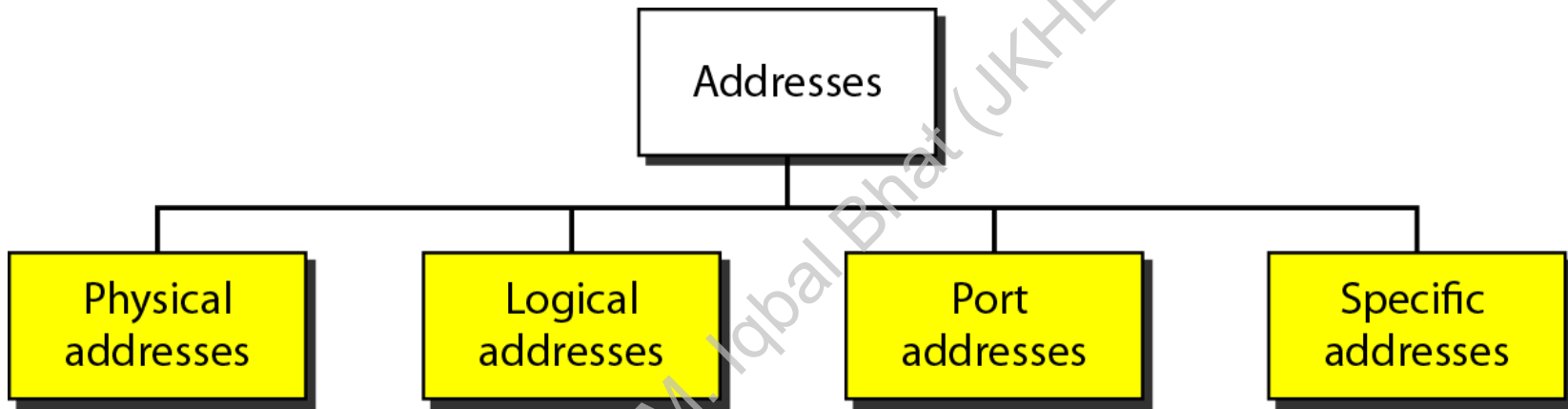


Figure 2.18 *Relationship of layers and addresses in TCP/IP*

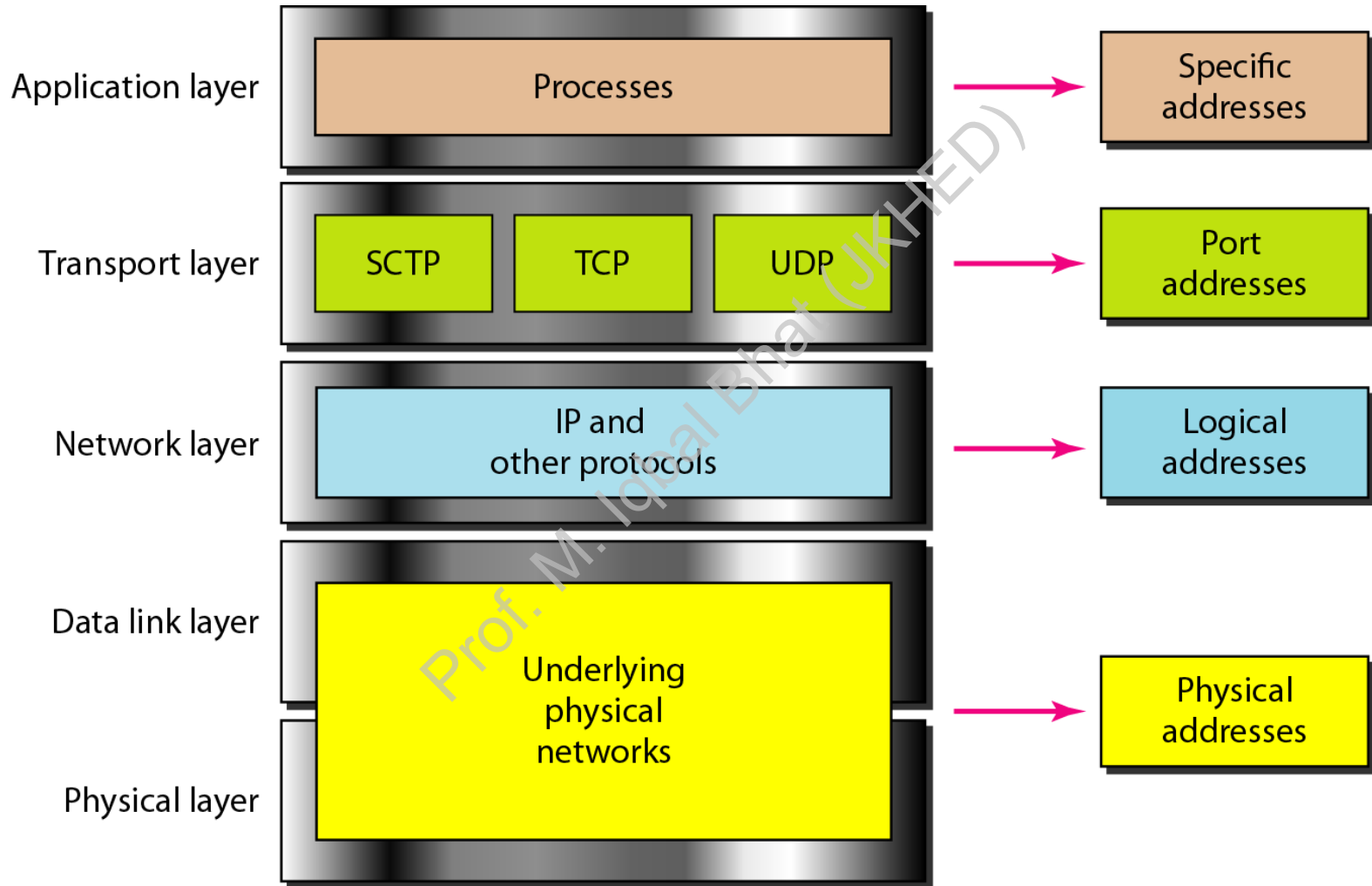


Figure 2.19 *Physical addresses*

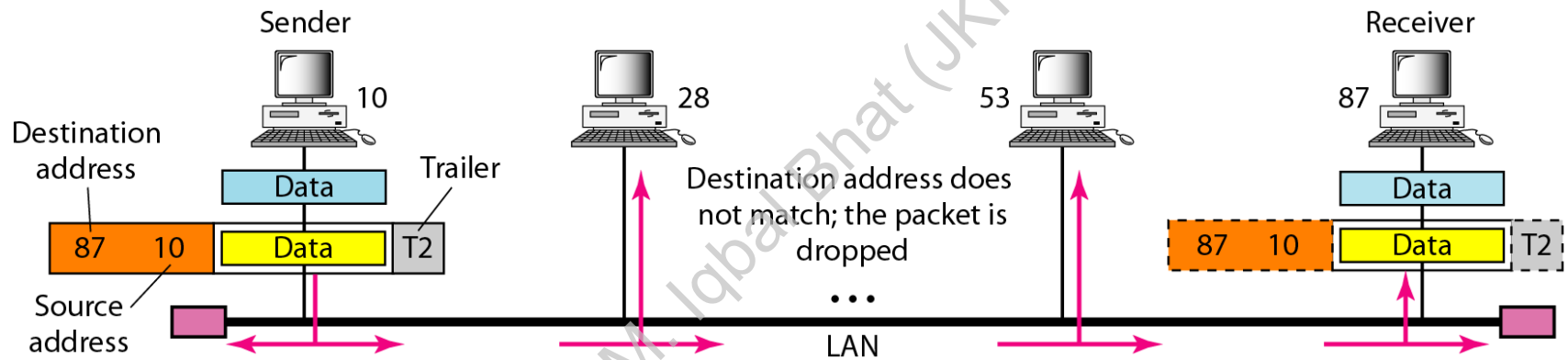


Figure 2.20 *IP addresses*

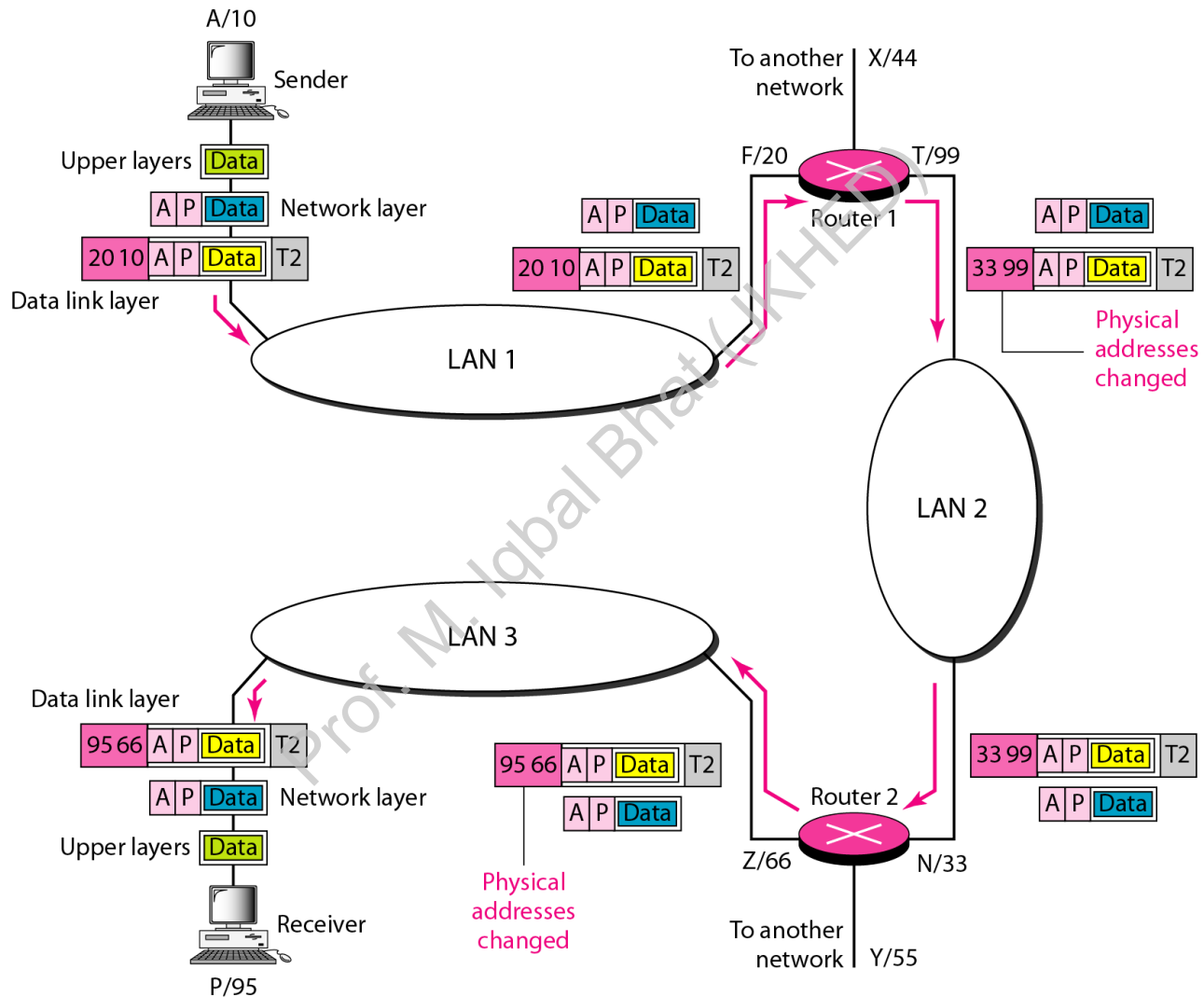
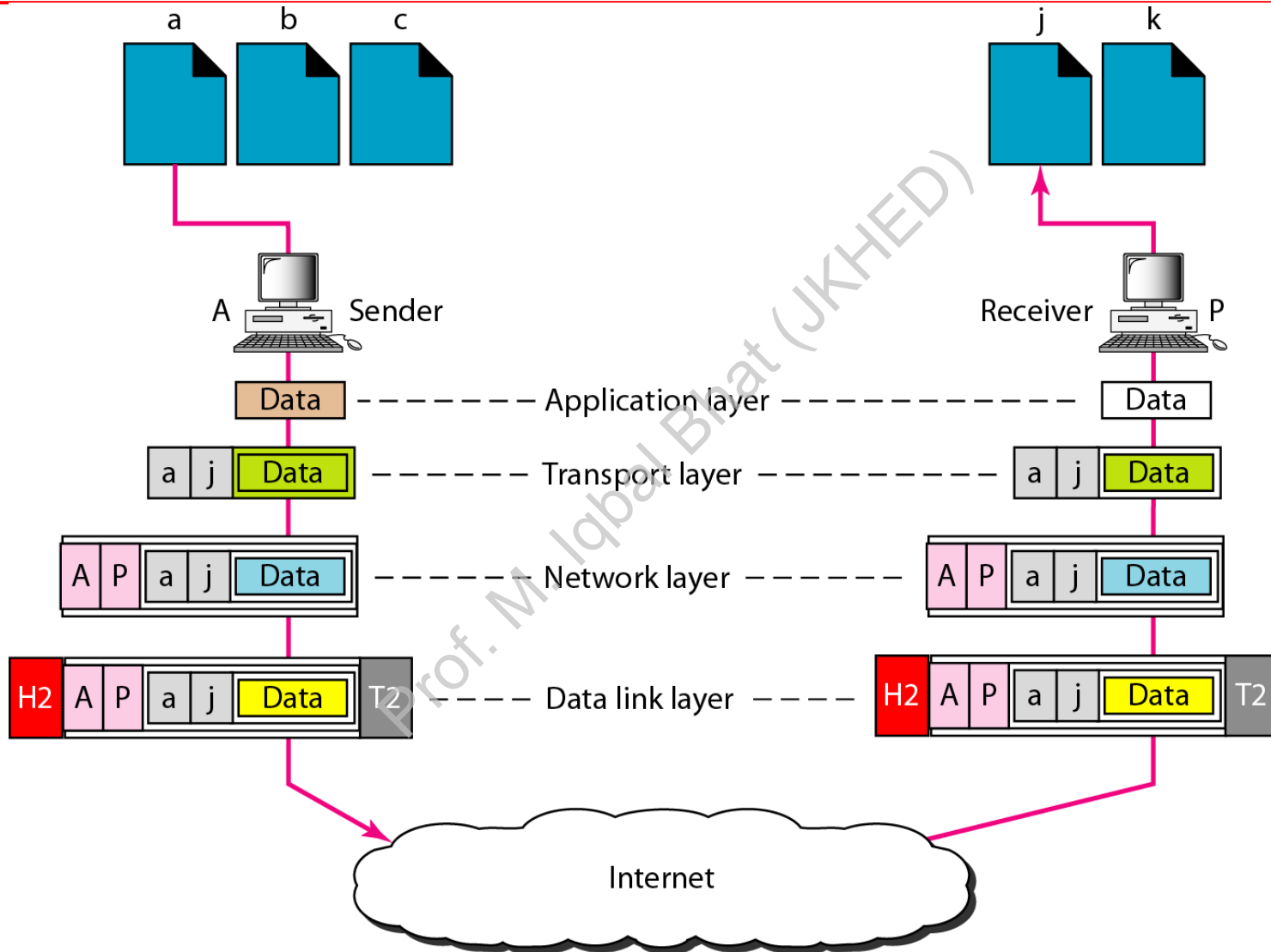


Figure 2.21 *Port addresses*



Note

**The physical addresses change from hop to hop,
but the logical and port addresses usually remain the same.**